Process Automation for the Aspirants and Contacts of the De La Salle Brothers Community of Lipa (Highlanders)

A Project Submitted to the College of Information Technology and Engineering

in partial fulfillment of the requirements

for the course

APPDEVT (Applications Development and Emerging Technologies)

Submitted by:

Cabauatan, Raymond Laurence

Crispospolo, Chanel Allyson

Marquez, Nicole Louise

Martinez, Tristan Mark

Pecayo, Miguel Alfonso

October 29, 2026

I. Company Profile

The Aspirants and Contacts of the De La Salle Brothers Community of Lipa, commonly known as the Highlanders, is a faith-based student organization established in 2017, currently having 116 members and 3 moderators (2 faculty members and 1 brother), all of whom serve on a voluntary, non-compensated basis. As a Lasallian formation organization, its nature of business is to give young Lasallians experiential exposure to the life, community, and vocation of the De La Salle Brothers through programs built around immersion, accompaniment, and discernment rather than classroom instruction alone. Its core activities include live-in immersions for college students, where students stay with a Brothers' community for a weekend to experience its daily rhythm of prayer, meals, ministry, and community life; and the Aspirancy program, where each aspirant is assigned a mentor, pairing students with a Brother for accompaniment to help them learn more about themselves and their discernment. Through these activities, the organization supports the personal, spiritual, and vocational formation of its members while promoting awareness of the Brothers' charism and mission within the Lasallian community.

II. Functional Area

The study will be conducted within the Highlanders organization, specifically its Secretariat and Executive Department, the functional area responsible for keeping track of activities, tracking the attendance of each member, and ensuring all requirements needed from the members have been supplied. This department operates in

coordination with the Lasallian Youth Ministry Office (LYMO) and the De La Salle Brothers Community of Lipa, whose Brothers serve as formators, hosts, and mentors for the organization's activities, and with the 3 moderators, who provide institutional oversight and link the organization to the wider Lasallian campus all around the Lasallian East Asian District (LEAD).

Moderators: Approve the waiver policy and ensure compliance with institutional and legal requirements, and serve as the final approvers of the UiPath-generated automated reports.

Executive Council: Set attendance requirements, identify members needing waivers, and monitor compliance.

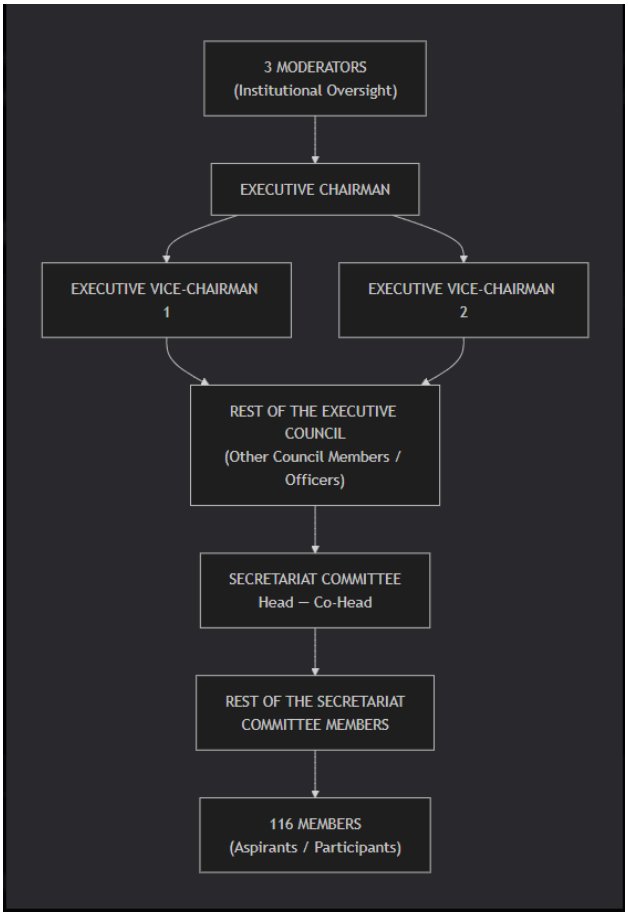
Secretariat Head & Co-Head: Assign tasks, maintain the master waiver list, and review submissions for completeness.

Secretariat Members: Record attendance, collect signed waivers, follow up on missing requirements, and encode records which serves as the primary data source.

Council Review: Check consolidated reports, flag incomplete submissions, and clear members for activities.

Moderators' Endorsement: Approve and archive the final attendance and waiver summary.

III. Organizational Chart



IV. Process Description

The proposed automation targets the attendance monitoring, recognition, waiver tracking, and reporting functions of the Highlanders, specifically the routine work currently performed manually by the Moderators and the Secretariat. Using UiPath, the system will handle repetitive data processing while leaving final verification to the Moderator. The following processes will be automated:

1. Attendance data ingestion from the existing Excel workbook
2. Attendance counting and ranking per member per week

3. Top Highlander of the Week identification
4. Weekly attendance report generation in PDF format
5. Waiver status checking per member
6. Report routing to the Moderator for review and verification

Step-by-Step Discussion of How the Processes Work

Step 1: Moderator Updates Excel (Manual)

The Moderator enters weekly attendance data into the Excel workbook. This remains manual to allow human judgment for excused absences or late entries.

Step 2: UiPath Reads Data

The robot opens the workbook, reads the relevant sheet, and validates that the file, sheet, and required columns exist. It logs an error and notifies the Moderator if the data is missing or malformed.

Step 3: System Calculates Attendance

The robot counts the present marks per member for the week based on org rules and stores the results in a data table for ranking.

Step 4: System Identifies "Top Highlander"

The robot sorts the counts to find the highest. It applies a tiebreaker rule (co-awardees) if necessary and tags the winner(s) for Moderator confirmation.

Step 5: System Checks Waivers

The robot cross-references members against the waiver tracking sheet, flags anyone with missing, expired, or unsigned waivers, and creates an exception list for the Secretariat.

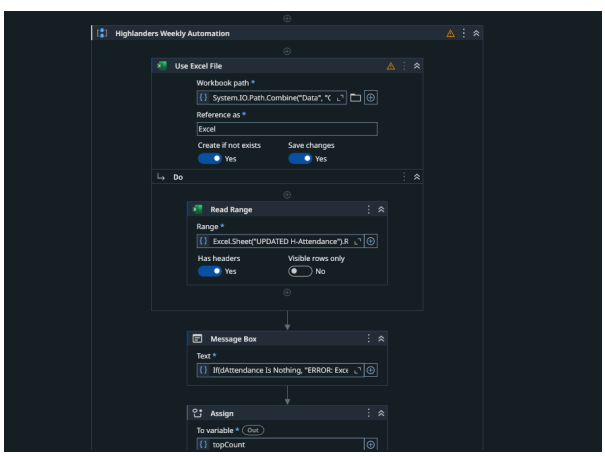
Step 6: System Generates PDF Report

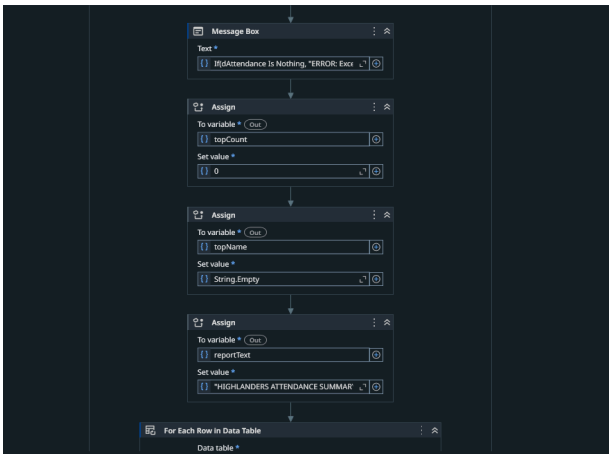
The robot creates a PDF report containing the attendance table, per-member counts, the Top Highlander(s), and summary statistics, saving it to a designated folder with a standardized filename.

Step 7: Moderator Reviews and Verifies

The robot emails the PDF report to the Moderator. The Moderator reviews the data, verifies the Top Highlander, and either approves the award email or requests corrections. Once approved, records are archived.

V. Screenshot





Use Excel File: The path is dynamically constructed using System.IO.Path.Combine(“”) ensuring the bot looks in the correct folder. It is referenced in the automation as Excel.

Read Range: Inside the Excel scope, the bot reads the specific sheet named "UPDATED H-Attendance".

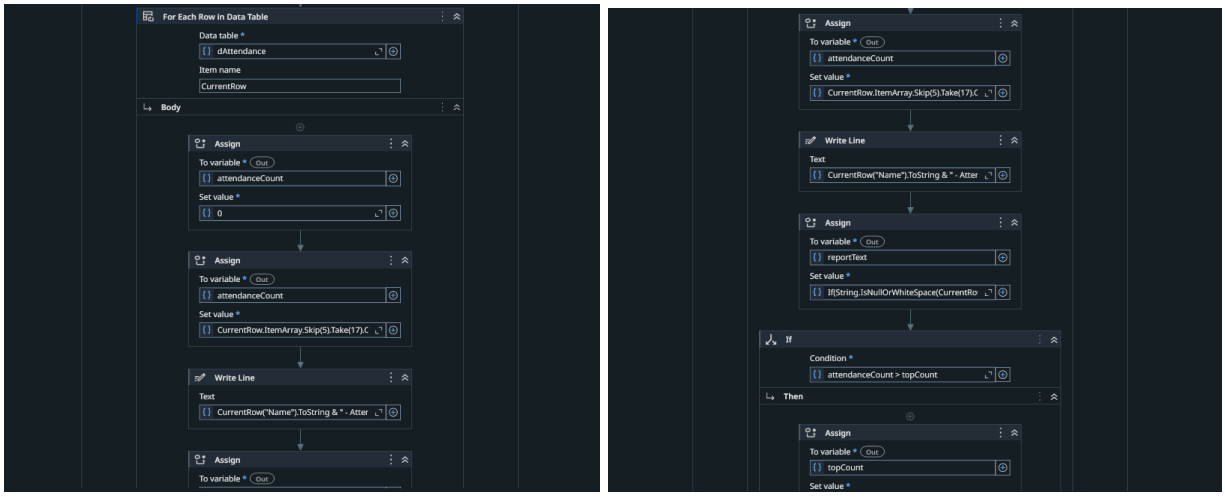
Message Box (Validation): The bot performs a quick validation check. This ensures the bot doesn't crash later in the process.

Assign (topCount): The variable topCount (Integer) is initialized to 0. This will be used to track the highest number of attendance marks.

Assign (topName): The variable topName (String) is initialized to String.Empty. This will store the name of the Top Highlander.

Assign (reportText): The variable reportText (String) is initialized. This acts as the starting header for the text report that will be generated.

For Each Row in Data Table: The bot enters a loop to process the attendance data row by row.



Assign (attendanceCount): Inside the loop, a variable attendanceCount is reset to 0 for each new row (member).

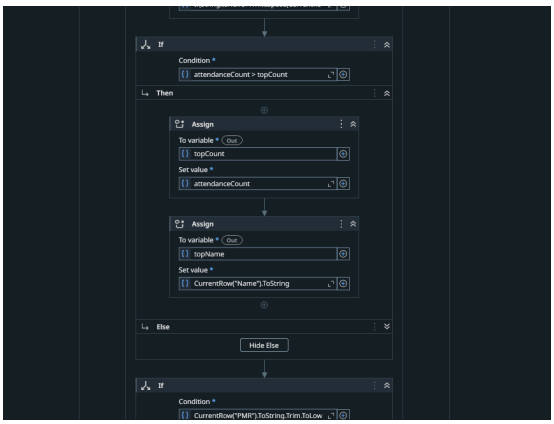
Assign (attendanceCount calculation): The bot uses LINQ (.ItemArray.Skip(5).Take(17)) to extract the attendance marks for the week. It skips the first 5 columns (likely ID, Name, P/MR status, etc.) and takes the next 17 columns (representing the days of the week). It then calculates the count.

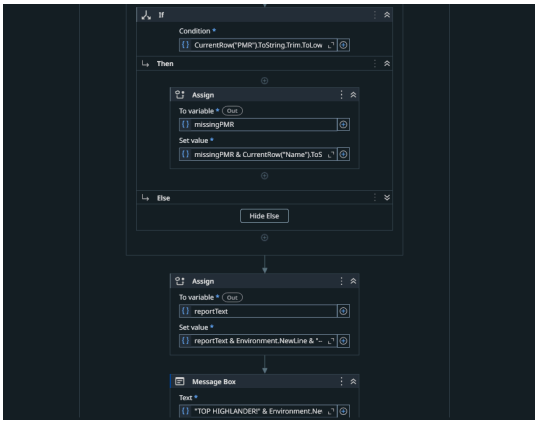
Write Line: A log is created to track the member's name and their calculated attendance count for debugging and monitoring purposes.

Assign (reportText formatting): The bot appends the current member's attendance summary to the reportText variable, ensuring a clean format for the final report.

If (Condition): The bot checks if attendanceCount > topCount. This is the logic to find the highest attendance.

Then (Assign topCount & topName): If the current member's attendance is higher than the previous highest (topCount), the bot updates topCount to the new number and updates topName to the current member's name.



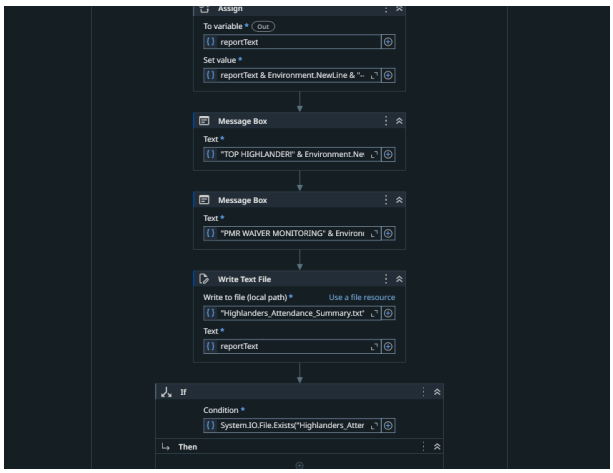


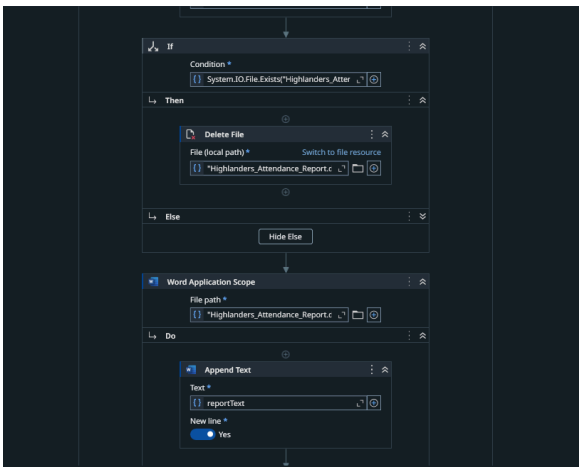
If (Condition): Still inside the loop, the bot checks the "PMR" column (which tracks waiver/medical requirements). It converts the text to lowercase and trims spaces, checking if it says "no" or is empty.

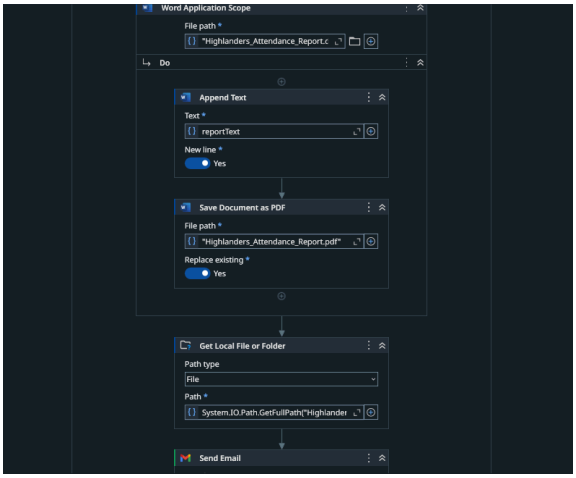
Then (Assign missingPMR): If the condition is true (meaning the member is missing a waiver), the bot appends their name to a missingPMR string variable.

Assign (reportText formatting): Outside the loop, the bot appends a visual divider (------) to the end of the attendance summary.

Message Box: A pop-up displays the name of the identified "TOP HIGHLANDER" and their count, allowing the user to see the result immediately.







Message Box (Waiver Monitoring): Another pop-up appears, displaying the list of members who are missing waivers (the missingPMR variable).

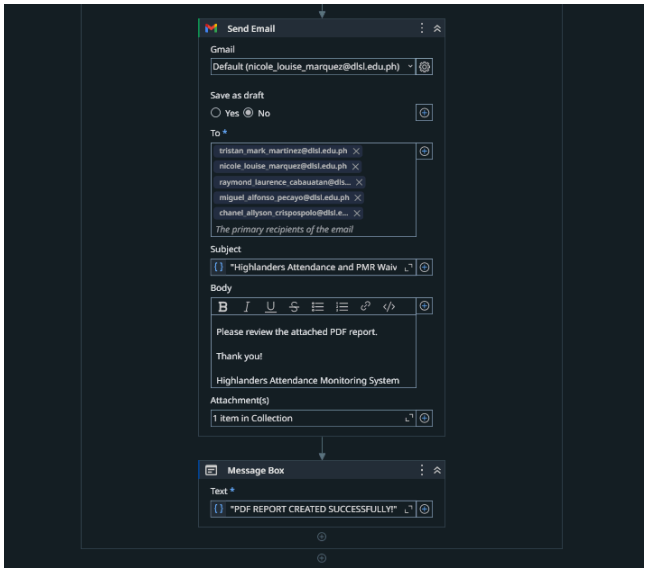
Write Text File: The bot generates a text file containing the reportText.

If (File Exists): The bot checks if an existing Word/PDF report file already exists.

Word Application Scope: The bot opens a Word document (likely a template). It uses the Append Text activity to paste the reportText generated earlier into the document.

Save Document as PDF: The Word document is immediately saved and exported as a PDF file named "Highlanders_Attendance_Report.pdf". "Replace existing" is set to Yes.

Get Local File or Folder: The bot fetches the full file path of the newly created PDF so it can be attached to an email.



Send Email: The bot uses a Gmail account to send the report.

- To: The email is routed to the Moderators (the @dlsl.edu.ph addresses).
- Subject: "Highlanders Attendance and PMR Waiver..."
- Body: A standard message: "Please review the attached PDF report. Thank you! Highlanders Attendance Monitoring System".
- Attachment: The PDF file generated in the previous step is attached.

Message Box (Completion): The final step of the script displays "PDF REPORT CREATED SUCCESSFULLY" to notify the user that the entire automated process is finished.